

A016 — Helmholtz Vorticity-Population / Matter–Field Partition Falsifier

Falsification Protocol and Final Report

SST-Workbench

Version v0.2.0

1 Identification, scope, and status

Field	Value
Catalog ID	A016
Falsifier	helmholtz_vorticity_population_matter_field_partition
Version	v0.2.0
Framework	SST Falsifier Framework v1.0.6 / SST-FALSIFIER-RUNTIME-2
Profile	cpu: Python FP64 reference plus strict C++/OpenMP FP64 certification
Status	Frozen protocol is created by <code>run_all.cmd FREEZE</code> ; later results are inserted automatically without editing this text.

This version preserves the numerical thresholds of A016 v0.1.1 and adds independent material-population and exterior-field tests. A physical/scientific gate failure does not terminate sibling scientific branches.

2 Research question and objective

Can a smooth incompressible inviscid vortex model simultaneously satisfy: (i) material preservation of the zero/nonzero vorticity population; (ii) material-volume and vorticity-flux conservation; (iii) a locally irrotational but nontrivial exterior velocity field with circulation periods and nonlocal kinetic energy; and (iv) the original static relaxed-centerline relative-equilibrium test?

The objective is not to prove an SST interpretation. It is to make the Helmholtz-to-SST partition testable with explicit observables, provenance, convergence, and backend certification. The static centerline branch cannot establish actual long-time Euler persistence or reconnection avoidance.

3 Competing hypotheses and allowed conclusions

Null hypothesis H_0 . At least one preregistered consequence fails beyond tolerance: the exact material control does not preserve the vorticity class/volume/flux, the reconstructed exterior is not locally potential, exterior kinetic energy is unresolved, the far-field exponent is unstable on eligible closed sources, or an input fails one of the unchanged v0.1.1 centerline gates.

Alternative hypothesis H_1 . The exact material controls close numerically; the exterior is locally divergence-free/curl-free but can retain nonzero speed and circulation periods; a resolved energy fraction lies outside the near-core source zone; eligible closed sources have stable far-field scaling; and each source geometry receives an independent H0–H4 classification.

Falsification semantics. Each H* or P* branch reports independently after G0/G1/G2. Aggregate FAIL means at least one eligible input failed that branch; per-input results remain in `BLIND_SAMPLE_METRICS.json`. Backend failure is certification evidence only. No threshold may be moved after freeze.

4 Assumptions, domain of validity, and known edge cases

- A1. Smooth three-dimensional incompressible Euler flow, no viscosity, no external nonconservative forcing.
- A2. Source geometries are closed oriented filaments. Finite-core regularization is used only where self-interaction/energy would otherwise be singular.
- A3. The blind geometry branch uses source units and unit circulation. No physical density or absolute length calibration participates in blind scoring.
- A4. P3 samples only exterior points. P5 uses a geometric near-core mask and therefore tests nonlocal velocity-energy support, not exact compact vorticity support.
- A5. P6 excludes near-zero-vector-area geometries because cancellation may remove the leading localized-source far-field moment.

The experiment does not cover viscosity, compressibility, reconnection, singular Euler breakdown, or externally forced vorticity production.

5 Symbols, units, and dimensional consistency

Symbol	Meaning	SI unit	Dimension
$\mathbf{x}$	position	m	L
t	flow time	s	T
$\mathbf{u}$	velocity	m s^{-1}	LT^{-1}
$\boldsymbol{\omega}$	vorticity $\nabla \times \mathbf{u}$	s^{-1}	T^{-1}
$\mathbf{F}$	deformation gradient	1	1
J	$\det \mathbf{F}$	1	1
Γ	circulation/vorticity flux	$\text{m}^2 \text{s}^{-1}$	$L^2 T^{-1}$
a	core radius	m after scale choice	L
ℓ_E	$E/(\rho \Gamma^2)$	m	L
p	velocity far-field exponent	1	1

All machine-used equations and dimensional checks are duplicated in the frozen `science_contract.json` and rendered below.

6 Exact mathematical model and observables

The central continuum relations are

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad (1)$$

$$\boldsymbol{\omega}(\mathbf{X}(\mathbf{a}, t), t) = \mathbf{F}(\mathbf{a}, t) \boldsymbol{\omega}_0(\mathbf{a}), \quad (2)$$

$$J = \det \mathbf{F} = 1, \quad (3)$$

$$\Phi_\omega = \int_{S(t)} \boldsymbol{\omega} \cdot d\mathbf{S} = \Gamma = \text{const.} \quad (4)$$

For the numerical filament reconstruction,

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{x} - \mathbf{x}')}{(|\mathbf{x} - \mathbf{x}'|^2 + a^2)^{3/2}}, \quad (5)$$

with $a = 0$ only for exterior evaluations that remain separated from the filament. The energy-length coefficient is

$$E = \rho \Gamma^2 \ell_E, \quad \ell_E = \frac{1}{8\pi} \oint \oint \frac{d\boldsymbol{\ell} \cdot d\boldsymbol{\ell}'}{\sqrt{|\mathbf{x} - \mathbf{x}'|^2 + a^2}}. \quad (6)$$

Relative equilibrium is tested after fitting rigid translation, rigid rotation, and tangential reparameterization. Exterior topology is tested through

$$\Gamma^{-1} \oint_{\gamma} \mathbf{u} \cdot d\boldsymbol{\ell} \approx \text{Lk}(\gamma, C). \quad (7)$$

The exterior differential audit tests $\nabla \cdot \mathbf{u} = 0$ and $\nabla \times \mathbf{u} = 0$. The energy partition observable is

$$f_{\text{ext}} = \frac{\int_{V \setminus V_{\text{core}}} \frac{1}{2} |\mathbf{u}|^2 dV}{\int_V \frac{1}{2} |\mathbf{u}|^2 dV}. \quad (8)$$

Far-field shells fit

$$\log u_{\text{rms}}(r) = b - p \log r. \quad (9)$$

A steady irrotational Bernoulli interpretation maps the frozen velocity exponent to $|\Delta p| \propto r^{-2p}$ and $|\nabla p| \propto r^{-(2p+1)}$; this is a post-fit interpretation, not a fitting constraint.

Machine-readable science contract

Research question Within smooth incompressible inviscid vortex dynamics, can a fixed material population carrying nonzero vorticity coexist with an exterior locally irrotational sector that still carries nonzero velocity, circulation periods, kinetic energy, and a reproducible far field, while the original A016 relaxed-centerline relative-equilibrium test remains independently falsifiable?

Objective Fuse the v0.1.1 Helmholtz centerline gates with theorem-level Cauchy-map controls and explicit matter-field-partition observables, without retuning the original H0-H4 thresholds and without allowing an early scientific FAIL to suppress later independent measurements.

H₀ At least one registered ideal-flow consequence or centerline condition fails beyond its preregistered tolerance: material zero/nonzero vorticity classification is not preserved by the tested incompressible map, material volume or vorticity flux is not conserved, the exterior is not numerically potential, exterior energy is unresolved or absent, far-field decay is not stable on eligible closed sources, or a relaxed centerline fails its original relative-equilibrium criterion.

H₁ The exact material controls satisfy Cauchy/volume/flux invariance, the reconstructed exterior is locally divergence-free and curl-free but nontrivial, a resolved nonzero kinetic-energy fraction lies outside the near-core source zone, eligible closed filaments show a stable localized-source far-field exponent, and each relaxed centerline receives an independent H0-H4 classification without threshold adjustment.

Registered equations

E1: Define vorticity

$$\boldsymbol{\omega} = \nabla \times \mathbf{u} \quad (10)$$

Dimensional check: curl contributes L^{-1} , so $[\boldsymbol{\omega}] = (L T^{-1}) L^{-1} = T^{-1}$.

E2: Cauchy vorticity map for incompressible Euler

$$\boldsymbol{\omega}(\mathbf{X}(\mathbf{a}, t), t) = \mathbf{F}(\mathbf{a}, t) \boldsymbol{\omega}_0(\mathbf{a}), \quad \mathbf{F} = \partial \mathbf{X} / \partial \mathbf{a} \quad (11)$$

Dimensional check: $\mathbf{F}$ is dimensionless and both vorticities have T^{-1} .

E3: Incompressible material-volume constraint

$$J = \det \mathbf{F} = 1 \quad (12)$$

Dimensional check: determinant of a dimensionless deformation gradient is dimensionless.

E4: Vorticity-flux/circulation conservation control

$$\Phi_\omega = \int_{S(t)} \boldsymbol{\omega} \cdot d\mathbf{S} = \Gamma = \text{const} \quad (13)$$

Dimensional check: T^{-1} times L^2 equals $L^2 T^{-1}$.

E5: Unit-circulation Biot-Savart reconstruction outside and on regularized filament fields

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{x} - \mathbf{x}')}{(|\mathbf{x} - \mathbf{x}'|^2 + a^2)^{3/2}} \quad (14)$$

Dimensional check: ΓL times L divided by L^3 gives $L T^{-1}$.

E6: Finite-core line energy coefficient

$$E = \rho \Gamma^2 \ell_E, \quad \ell_E = \frac{1}{8\pi} \oint \oint \frac{d\boldsymbol{\ell} \cdot d\boldsymbol{\ell}'}{\sqrt{|\mathbf{x} - \mathbf{x}'|^2 + a^2}} \quad (15)$$

Dimensional check: ℓ_E has L ; $\rho \Gamma^2 \ell_E$ gives $(M L^{-3})(L^4 T^{-2})L = M L^2 T^{-2}$.

E7: Relative-equilibrium decomposition

$$\mathbf{u}_{\text{ind}}(\mathbf{X}) = \mathbf{U} + \boldsymbol{\Omega} \times (\mathbf{X} - \mathbf{X}_0) + \lambda(s)\mathbf{t}(s) + \mathbf{r}_\perp(s) \quad (16)$$

Dimensional check: each term has velocity dimensions $L T^{-1}$; λ has $L T^{-1}$.

E8: Exterior circulation period and linking consistency

$$\frac{1}{\Gamma} \oint_\gamma \mathbf{u} \cdot d\boldsymbol{\ell} \approx \text{Lk}(\gamma, C) \in \mathbb{Z} \quad (17)$$

Dimensional check: line integral has $L^2 T^{-1}$ and division by Γ is dimensionless.

E9: Local exterior potential-flow audit

$$\nabla \cdot \mathbf{u} = 0, \quad \nabla \times \mathbf{u} = \mathbf{0} \quad \text{away from vortex support} \quad (18)$$

Dimensional check: both divergence and curl of velocity have T^{-1} .

E10: Exterior kinetic-energy fraction proxy

$$f_{\text{ext}} = \frac{\int_{V \setminus V_{\text{core}}} \frac{1}{2} |\mathbf{u}|^2 dV}{\int_V \frac{1}{2} |\mathbf{u}|^2 dV} \quad (19)$$

Dimensional check: identical energy-density-volume factors cancel, leaving a dimensionless fraction.

E11: Far-field localized-source exponent estimator

$$\log u_{\text{rms}}(r) = b - p \log r, \quad p = -\frac{d \log u_{\text{rms}}}{d \log r} \quad (20)$$

Dimensional check: the slope of logarithms of consistently scaled quantities is dimensionless.

E12: Post-reveal Bernoulli exponent map for a steady irrotational exterior

$$|\Delta p| \propto r^{-2p}, \quad |\nabla p| \propto r^{-(2p+1)} \quad (21)$$

Dimensional check: This equation maps only exponents; dimensional pressure normalization is not inferred by the blind fit.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol and resolve the exact source directory from `source_contract.json` plus `SST_WORKBENCH_ROOT`.
2. **S2** [G2; equations E5, E6]: Run self-contained Python FP64 circle sanity controls before external source scoring.
3. **S3** [P0; equations E2]: Evaluate the exact diagonal incompressible Cauchy map for zero/nonzero vorticity populations.
4. **S4** [P1; equations E3]: Evaluate $\det(\mathbf{F})$ and the analytic material cross-section scaling.
5. **S5** [P2; equations E4]: Evaluate initial/final vorticity flux under the same exact stretch.
6. **S6** [H1; equations E5, E6, E7, E8]: Resample each closed source geometry, estimate finite-core thickness, and calculate original A016 H0-H4 observables at preregistered resolutions.
7. **S7** [P3; equations E5, E9]: Sample an exterior sphere and use centered finite differences of singular-kernel Biot-Savart velocity to estimate divergence and curl.
8. **S8** [P4; equations E8, E9]: Reuse meridian circulation periods together with nonzero exterior speed to test locally irrotational but globally nontrivial exterior flow.
9. **S9** [P5; equations E5, E10]: Integrate the finite-core velocity-energy proxy with deterministic Halton-ball samples at two sample counts and partition by the preregistered near-core mask.
10. **S10** [P6; equations E11, E12]: Fit log RMS speed versus log radius on preregistered far-field shells for nonzero-vector-area closed sources.
11. **S11** [B0; equations E5]: Build the instance-local C++17/OpenMP kernel and compare its FP64 Biot-Savart output against the Python FP64 reference with no fallback authority.

7 Exact algorithm: equation-to-code mapping

- S1. Verify the frozen protocol and resolve the source path only from `source_contract.json` and `SST_WORKBENCH_ROOT`.
- S2. Execute self-contained Python FP64 circle controls.
- S3. Execute the exact diagonal incompressible affine Cauchy-map control for P0–P2.
- S4. Parse each centerline, remove a duplicated terminal point, and arclength-resample to the registered resolutions.
- S5. Compute the unchanged H0–H4 geometry, convergence, holonomy, relative-equilibrium, and symmetry observables.
- S6. Sample an exterior sphere and estimate divergence/curl by centered finite differences of the singular-kernel exterior field.
- S7. Estimate exterior kinetic-energy fraction with deterministic Halton-ball samples at two counts.
- S8. Fit far-field RMS speed on fixed radius factors for vector-area-eligible geometries.
- S9. Build the instance-local C++17/OpenMP FP64 kernel and compare it with the Python FP64 reference.
- S10. Freeze all blind results before the framework permits reveal.

The gate graph is a fan-out graph after infrastructure prerequisites; it is deliberately not a serial scientific pass chain.

8 Source, provenance, and independence contract

The admissible external source is `KnotPlot/knots/final` under the resolved Workbench root. One file is one sampling unit. Resolutions of the same file, mirrored versions, and backend repeats are not independent units. The parser accepts one or more XYZ components separated by blank lines. Exclusion occurs only for parse/geometry failure or preregistered P6 vector-area ineligibility.

9 Gate DAG and decision semantics

The infrastructure spine is $G0 \rightarrow G1$ and $G0 \rightarrow G2$. H0–H4 and P3–P6 require G0/G1/G2 but do not require each other. P0–P2 and B0 require G0/G2. Thus, for example, H3=FAIL does not prevent P3–P6 or B0 from running.

Gate	Question	PASS	FAIL/other
G0	frozen protocol integrity	verified	process stop before pipeline
G1	source admissibility	source directory exists	source FAIL; source-dependent gates are not run
G2	Python reference controls	finite, positive energy; reversal error $\leq 10^{-12}$	numerical control FAIL
H0	closed geometry/thickness	$N \geq 32$, closure ratio ≤ 2.5 , thickness/edge ≥ 1.25	any eligible input fails

H1	convergence	energy and RE drift within 0.08	any eligible input fails
H2	holonomy/linking	at least two valid loops and max error ≤ 0.08	any eligible input fails
H3	relative equilibrium	normal NRMSE ≤ 0.18	any eligible input fails
H4	covariance	both errors $\leq 10^{-9}$	any eligible input fails
P0	vorticity population	Cauchy error $\leq 10^{-12}$ and no class switch	exact control fails
P1	material volume	$ J - 1 \leq 10^{-12}$	exact control fails
P2	vorticity flux	relative drift $\leq 10^{-12}$	exact control fails
P3	exterior potentiality	valid points ≥ 8 ; div/curl residuals ≤ 0.08	any eligible input fails
P4	exterior circulation	H2-valid period plus nonzero exterior speed	any eligible input fails
P5	exterior energy	$f_{\text{ext}} \geq 0.01$; refinement drift ≤ 0.12	any eligible input fails
P6	far-field exponent	$R^2 \geq 0.97$, $2.4 \leq p \leq 3.6$ on eligible inputs	any eligible input fails; none eligible is unresolved
B0	C++ certification	relative L2 $\leq 10^{-10}$	build/parity certification FAIL
G9	reveal commitment	framework verification	reveal denied if policy fails

10 Numerical certification, convergence, and error budget

BASIC uses $N = 96,160$; FULL uses $N = 160,256$; CERTIFY uses $N = 256,384$. H1 compares finite-core energy and relative-equilibrium residual between the two registered resolutions. Holonomy uses at least 384 source points. P5 uses 1024 and 4096 deterministic Halton-ball samples. P3 uses an exterior centered-difference step $h = 0.002R$. P6 uses radius factors 4, 6, 8, 12 times the source bounding radius.

Error source	Estimator	Acceptance
Centerline discretization	H1 relative differences	≤ 0.08 for registered energy/RE measures
Holonomy quadrature	integer absolute error	≤ 0.08
Exterior finite difference	dimensionless div/curl residual	≤ 0.08
Energy sampling	low/high f_{ext} drift	≤ 0.12
Far-field fit	linear-log R^2	≥ 0.97
Backend round-off/implementation	C++ vs Python relative L2	$\leq 10^{-10}$

11 Backend authority and parity

Python/NumPy FP64 is REFERENCE authority. The instance-local C++17 kernel is built through Framework v1.0.6 and receives CERTIFICATION authority only when the actual backend is serial or OpenMP FP64 and parity passes. A Python fallback cannot close B0. GPU screening is not enabled in this A016 version.

12 Blindness, opaque identifiers, and reveal

Input filenames are replaced in public sample metrics by 16-hex HMAC-SHA256 identifiers derived from the file hash and a private key. The runtime identity map is stored under the private output subtree, excluded from the blind ZIP. Target-bearing assignment terms listed in the private forbidden-term policy are commitment-protected. Reveal is permitted after FULL or CERTIFY when G2 passed; unresolved scientific branches do not block reveal because the design intentionally preserves informative later measurements after earlier scientific failures.

13 Statistical/null model and multiplicity

This is primarily a deterministic numerical falsifier, not a parameter-fitting study. The independent unit is one source file. H*/P* aggregate status is conservative: any eligible per-input failure makes the aggregate branch FAIL. P6 eligibility is fixed by vector-area ratio before exponent acceptance. P5 sampling uses deterministic low-discrepancy points rather than stochastic confidence intervals. No multiplicity correction is used because there is no null-hypothesis p-value family; all per-input values and thresholds are reported directly.

14 Results and automatic run provenance

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	17158e263151546885b66290e74ba6f0a6fbb456c6a3303dc0141545b20c586c
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	3f54e072e255e72c85813833f906e677ebf395a528003f6a0e9c07886a08e3a4

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen v1.0.6 protocol intact at runtime?
G1	FAIL	Is the preregistered relaxed-centerline source present and provenance-
G2	PASS	Do analytic and numerical controls validate the Python FP64 referenc
P0	PASS	Does the Cauchy vorticity map preserve the zero/nonzero vorticity po
P1	PASS	Does the same material map conserve material volume to numerical t
P2	PASS	Does vortex stretching preserve vorticity flux through the correspondi
H0	NOT_RUN_PREREQUISITE	Do input centerlines satisfy closed-filament geometry and finite-thickn
H1	NOT_RUN_PREREQUISITE	Are finite-core energy and relative-equilibrium observables resolution-
H2	NOT_RUN_PREREQUISITE	Do exterior meridian circulation periods agree with integer linking str

H3	NOT_RUN_PREREQUISITE	Are the relaxed centerlines relative equilibria modulo translation, rotation, and reflection
H4	NOT_RUN_PREREQUISITE	Do circulation reversal and mirror transformations satisfy the register
P3	NOT_RUN_PREREQUISITE	Is the velocity field outside the vortex support numerically divergence-free
P4	NOT_RUN_PREREQUISITE	Does the irrotational exterior retain nontrivial circulation periods link
P5	NOT_RUN_PREREQUISITE	Is a resolved nonzero fraction of kinetic energy carried outside the pre
P6	NOT_RUN_PREREQUISITE	Does the closed-filament far field exhibit a stable localized-source deca
B0	PASS	Does strict C++/OpenMP FP64 reproduce the Python FP64 Biot-Savart
G9	DEFERRED	After the blind analysis, does commitment verification permit reveal a

Backend manifest

```
{
  "continue_after_scientific_fail": true,
  "cpp_certification": {
    "actual_backend": "openmp",
    "authority": "CERTIFICATION",
    "build_fingerprint_sha256": "58b8508c30ce50af6cac0d394a183fcb82640bb0e5cfa945c5992840ca320c",
    "device_name": null,
    "metrics": {
      "backend": "openmp",
      "openmp_compiled": true,
      "openmp_max_threads": 20,
      "precision": "float64"
    },
    "ok": true,
    "precision": "float64",
    "requested_backend": "cpp",
    "source_sha256": "b5fb8698c65a21a71e7a731dac0d42202c2d4b7f1ffe6a76d99acbb8b5838da1"
  },
  "n_public_samples": 0,
  "n_sample_errors": 0,
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "AO16-BACKEND-MANIFEST-2"
}
```

Source manifest

```
{
  "manifest_sha256": "91b9aad86af9ba8e8b7aa58b403f02d0ac043f1ff717a31938d5e27c448a4bf1",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "KNOTPLOT_RIDGERUNNER_FINAL",
      "evidence_class": "simulation",
      "exists": false,
      "independence_family": "relaxed-centerline-library",
      "path": "C:\\workspace\\projects\\SST-Workbench\\KnotPlot\\knots\\final",
      "provenance_family": "KnotPlot/RidgeRunner relaxed geometries",
      "registry_hint": {
```

```

        "catalog_id": "KNOTPLOT_RIDGERUNNER_FINAL",
        "hits": []
    },
    "role": "discovery",
    "source_id": "SO_RELAXED_CENTERLINES",
    "version": "local-workbench-manifested"
}
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

The frozen text above is never rewritten to accommodate an outcome.

15 Interpretation, limits, falsifiable predictions, and minimal follow-up experiments

P0–P2 test the exact material-population logic of the smooth incompressible model, not a full time-integrated knot evolution. H3 remains the original strong static-centerline shape test and can fail while the matter–field partition branch remains informative. P3/P4 establish whether the computed exterior is locally potential yet globally circulation-bearing. P5 tests whether velocity-energy support extends beyond the near-core source zone. P6 supplies a frozen far-field exponent for later mechanism comparison.

A minimal next experiment after v0.2.0 is a genuinely time-dependent Euler/finite-core evolution in which material tracers are initialized both inside and outside a vortex tube, then tested for support-class preservation, circulation conservation, and topology change under simultaneous spatial and temporal refinement. A second independent experiment should reconstruct pressure from the time-dependent velocity field rather than infer pressure solely from a steady exterior relation.

16 Reproducibility and exact commands

Place the instance under the A016 Workbench family and keep the frozen framework under 06_templates/SST_Fal. If needed, set SST_WORKBENCH_ROOT. Then run:

```

run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd FULL
run_all.cmd CERTIFY
run_all.cmd REVEAL

```

The first three scientific modes write blind packages; reveal is commitment-verified and separately packaged.

17 References

References

- [1] H. von Helmholtz (1858), “Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen,” *Journal für die reine und angewandte Mathematik* **55**, 25–55. DOI: <https://doi.org/10.1515/crll.1858.55.25>.
- [2] A.-L. Cauchy (1815), “Théorie de la propagation des ondes à la surface d’un fluide pesant d’une profondeur indéfinie,” historical source for the Lagrangian vorticity invariants; modern formulation summarized in standard vortex-dynamics texts.

- [3] H. K. Moffatt (1969), “The degree of knottedness of tangled vortex lines,” *Journal of Fluid Mechanics* **35**, 117–129. DOI: <https://doi.org/10.1017/S0022112069000991>.
- [4] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press. DOI: <https://doi.org/10.1017/CB09780511624063>.
- [5] G. K. Batchelor (1967), *An Introduction to Fluid Dynamics*, Cambridge University Press. DOI: <https://doi.org/10.1017/CB09780511800955>.